

Chapter 4

Artifact: constant-envelope upper semicontinuity

A tool for the limit passage inside the risk measure

The failure point (F3) is the passage to the limit inside the risk measure: on a general probability space, a convex risk measure need not be upper semicontinuous along almost surely convergent sequences. The next lemma restores exactly this passage for OCE risk measures, provided all objectives share a deterministic lower envelope. It is the tool used in the proof of the approximation theorem.

Lemma 4.1. *Let $\ell : \mathbb{R} \rightarrow \mathbb{R}$ be convex and non-decreasing and let ρ_ℓ be the associated OCE. Let $(Y_m)_{m \in \mathbb{N}}$ and Y be random variables with*

$$Y_m \geq -B \text{ } \mathbb{P}\text{-a.s. for all } m,$$

for some deterministic $B \geq 0$. If $\liminf_{m \rightarrow \infty} Y_m \geq Y$ $\mathbb{P}$ -a.s., then

$$\limsup_{m \rightarrow \infty} \rho_\ell(Y_m) \leq \rho_\ell(Y).$$

Proof. Fix $w \in \mathbb{R}$. Since $-(Y_m + w) \leq B + |w|$ and ℓ is non-decreasing,

$$g_w := \ell(B + |w|) \geq \ell(-(Y_m + w)),$$

so g_w is a deterministic (hence integrable) constant dominating $\ell(-(Y_m + w))$. Extend ℓ to $[-\infty, \infty)$ by $\ell(-\infty) := \lim_{t \rightarrow -\infty} \ell(t)$; the extension is continuous and non-decreasing. For any real sequence (a_m) with $\sup_m a_m < \infty$ one has $\limsup_m \ell(a_m) \leq \ell(\limsup_m a_m)$. Applying this with $a_m = -(Y_m + w)$ and using $\limsup_m (-(Y_m + w)) = -\liminf_m (Y_m + w) \leq -(Y + w)$ a.s. yields

$$\limsup_{m \rightarrow \infty} \ell(-(Y_m + w)) \leq \ell(-(Y + w)) \text{ } \mathbb{P}\text{-a.s..}$$

Reverse Fatou applied to the sequence

$$g_w \geq \ell(-(Y_m + w))$$

gives

$$\limsup_{m \rightarrow \infty} \mathbb{E}[\ell(-(Y_m + w))] \leq \mathbb{E}[\ell(-(Y + w))].$$

Since $\rho_\ell(Y_m) \leq w + \mathbb{E}[\ell(-(Y_m + w))]$, one obtains $\limsup_m \rho_\ell(Y_m) \leq w + \mathbb{E}[\ell(-(Y + w))]$ for every w , and taking the infimum over w completes the proof. $\square$